

Infer&Fuse: An Effective Decompose-then-Compose Framework for Long-Text-to-Image Generation

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Abstract

Despite recent advancements, state-of-the-art text-to-image (T2I) diffusion models struggle with complex and semantically dense texts, often resulting in missing objects and attribute misalignment. To address the challenge in long-text-to-image (LT2I) generation, we propose Infer&Fuse, an effective Decompose-then-Compose (DtC) framework designed to empower T2I models with robust LT2I generation capabilities. Under the DtC paradigm, Infer&Fuse approximates the complex distribution conditioned on the long text as a composition of multiple distributions derived from simpler texts. Specifically, it first decomposes the long text into several sub-prompts for independent score calculation using the underlying T2I model. Subsequently, these scores are combined using respective guidance factors to construct a holistic concept. Since the contribution of each sub-prompt varies across spatial regions, Infer&Fuse introduces spatially-varying, pixel-wise *guidance maps* to facilitate score composition, ensuring distinct concepts are generated within the appropriate regions. Our theoretical analysis proves that Infer&Fuse achieves an effective scheme for score composition, guaranteeing the generated image captures the diverse semantics presented in the input. In addition, we integrate the composite score with the score solely derived from the long text, using a well-designed *blending map*. This incorporation of global information further enhances the visual coherence and semantic fidelity of the generated image. Extensive experiments demonstrate that diverse T2I models equipped with Infer&Fuse achieve consistent performance gains across multiple public benchmarks, outperforming previous methods in LT2I generation.

Introduction

Recent text-to-image (T2I) diffusion models (Rombach et al. 2022; Podell et al. 2023; Labs 2024; Wu et al. 2025a) are capable of generating high-quality visual content that faithfully aligns with the user-provided instruction. However, most of them fall short in handling complex and semantically dense texts. As shown in Fig. 1, even some advanced methods suffer from missing objects and attribute confusion in this challenging scenario, resulting in poor performance in semantic alignment (Jiao et al. 2025; Ghosh, Hajishirzi, and Schmidt 2023; Huang et al. 2025).

As mentioned in previous studies (Huang, Lin, and Du 2025; Yang et al. 2024), the challenge in long-text-to-image (LT2I) generation fundamentally stems from the scarcity

of training data with long text annotations and the fixed context window of the text encoders. For example, early methods (Podell et al. 2023; Rombach et al. 2022) using CLIP (Radford et al. 2021) only process 77 tokens, truncating descriptions and omitting the details. Additionally, there is an inherent bottleneck in representing dense semantic content within limited image tokens (Rassin et al. 2023), leading to missing or misaligned semantics.

Recently, a growing group of methods aims to enhance the LT2I generation capabilities. To mitigate the inefficiency of the inherent text encoder, some studies (Liu et al. 2025; Hu et al. 2024; Wu et al. 2025b) additionally introduce powerful Large Language Models (LLMs) (Raffel et al. 2020; Touvron et al. 2023; Bai et al. 2025) and fine-tune the pre-trained T2I models in their curated datasets with long text annotations. However, they suffer from significant training overhead while exhibiting poor generalization ability. Other approaches (Rassin et al. 2023; Chefer et al. 2023) optimize the noisy latents at each timestep using the tailored objectives to enhance each semantic element in the long text, such as increasing their attention scores. Nevertheless, they suffer from cumbersome per-step optimization and struggle to model abstract concepts (e.g., relations among objects), leading to visually rigid and incoherent results.

To alleviate the challenge, we introduce **Infer&Fuse**, an efficient Decompose-then-Compose (DtC) framework designed for empowering the LT2I generation capabilities of the T2I models. Formally, Infer&Fuse first applies “Decompose” operator to break the long text into several simple sub-prompts using a powerful LLM (Yang et al. 2025), where each sub-prompt encodes distinct semantic content and can be effectively handled by the pre-trained T2I model. Then, the “Compose” operator in Infer&Fuse combines the individual semantics from each sub-prompt, thereby constructing a holistic concept aligned with the long text. To achieve this, Infer&Fuse calculates the score function of each sub-prompt using the underlying T2I model, while combining them to guide the model to sample from a semantic-rich distribution. Since the contribution of each sub-prompt differs across spatial regions, Infer&Fuse constructs spatially-varying, pixel-wise *guidance maps* to modulate the score composition effectively. Compared to other alternative strategies, our method achieves an effective score composition by maximizing the posterior probability with respect to the long text. However,

A high angled partial view of a colorful puzzle with animals, insects, plants, and flowers on it. The background of the image features a vibrant garden scene with colorful flowers, green foliage, and a variety of insects, creating a lively and natural setting... The top left portion of the puzzle shows a group of red, yellow, and orange colored... On the mid and bottom of the puzzle is a brown or dark tan colored pot with a Hedgehog, which is an animal with white and brown color, sitting and featuring spines, small ears, black eyes, paws, and a curious expression, poking out and staring outwards. On the pot are two bees and a moth. A small owl, carved from real wood, sits on the right side of the puzzle... Behind the Owl, on the right side of the puzzle, appears to be a wooden...



Figure 1: Long-Text-to-Image (LT2I) generation. The input long text is sourced from DetailMaster (Jiao et al. 2025). As shown in the figure, most of the advanced methods (Esser et al. 2024; Yang et al. 2024; Sueyoshi and Matsubara 2024) exhibit poor performance, resulting in missing objects or attribute misalignment.

relying solely on guidance derived from sub-prompts entails a loss of global context, such as inter-dependencies among entities, leading to visual incoherence. To mitigate this, we further integrate the composite score with the score derived from the original long text using a designed *blending map*, incorporating the global guidance. Essentially, this map complements regions of weak sub-prompt guidance with global information, thereby enhancing visual coherence and recovering missing details. Benefiting from this, Infer&Fuse further enhances the image quality and fidelity.

Consequently, these designs distinguish our approach from previous methods (Wu et al. 2025b; Yang et al. 2024; Liu et al. 2022; Zhao et al. 2023), enabling superior performance on LT2I tasks. In summary, our main contributions are:

- We propose Infer&Fuse, a training-free method building upon the DtC framework, enhancing the LT2I generation capabilities of the T2I models.
- The proposed Infer&Fuse leverages spatially-varying, pixel-wise guidance maps to combine the scores derived from each sub-prompt, achieving an efficient score composition for sampling from a semantic-rich distribution defined by the input long text.
- In addition, Infer&Fuse incorporates the global guidance through a pixel-wise blending map, enhancing the visual coherency and semantic fidelity.
- We implement our Infer&Fuse upon different T2I models. The experimental results in public benchmarks demonstrate that our method outperforms in the LT2I generation task regardless of the underlying model architectures.

Related Works

Text-to-Image Generation. Text-to-Image (T2I) models aim to generate images that faithfully reflect the semantics specified in user instructions. Early diffusion models (Podell

et al. 2023; Rombach et al. 2022) estimate and remove the noisy from the current latent. Recently, the flow matching models (Esser et al. 2024; Lipman et al. 2022) predict the velocity field to derive the noisy latent for the subsequent step. Recent methods (Labs 2024; Esser et al. 2024) leverage DiT (Peebles and Xie 2023) architectures and employ more advanced text encoders (Raffel et al. 2020; Wang et al. 2024), achieving significant leaps in visual quality. Despite promising results, existing T2I models still struggle to efficiently process complex and semantically dense texts, often resulting in semantic omissions or confusion.

Challenges in Long-Text-to-Image Generation. Despite the significant advancement in T2I generation (Rombach et al. 2022; Podell et al. 2023; Esser et al. 2024), existing methods (Li et al. 2023; Gani et al. 2023) exhibit sub-optimal performance in long-text-to-image (LT2I) generation tasks. The fundamental challenge stems from the pronounced scarcity of large-scale LT2I datasets featuring rich long-text annotations. Constrained by this, most methods demonstrate poor generalization ability when processing long text inputs. Another challenge arises from the fixed context window of the widely-used text encoders (Radford et al. 2021), leading to the omission of semantic details (Huang, Lin, and Du 2025; Yang et al. 2024). Additionally, generating the dense semantics within a limited image tokens remains a challenge, resulting in object omission and attribute misalignment.

Method: Infer&Fuse

Decompose-then-Compose Framework

Given a long text input P_{full} , our goal is to generate the final image z_0 that faithfully reflects all semantics specified in P_{full} . Before delving into the details, we first formalize the Decompose-then-Compose (DtC) framework, which defines

the following two operators:

$$\begin{aligned} \{P_i\}_{i=1}^{N_c} &= \text{Decompose}(P_{\text{full}}), \\ \mathbf{z}_0 &= \text{Compose}(p_\theta(\cdot|P_i), \{P_i\}_{i=1}^{N_c}), \end{aligned} \quad (1)$$

where $\{P_i\}_{i=1}^{N_c}$ denotes the semantic elements obtained by decomposing P_{full} via the ‘‘Decompose’’ operator. Specifically, each P_i only encodes a distinct semantic subset of P_{full} , making it compatible with the pre-trained T2I models. For example, StructureDiffusion (Feng et al. 2022) employs the extracted noun phrases as P_i .

Then, the ‘‘Compose’’ operator leverages the T2I model p_θ to create the final image $\mathbf{z}_0$ using $\{P_i\}_{i=1}^{N_c}$. A promising approach (Liu et al. 2022) to implement this operator is simulating the target score function $s_t^\theta(\mathbf{z}_t|P_{\text{full}}) = \nabla_{\mathbf{z}_t} \log(p_\theta(\mathbf{z}_t|P_{\text{full}}))$ through a series of individual ones $s_t^\theta(\mathbf{z}_t|P_i)$. Specifically, most previous studies (Liu et al. 2022; Zhao et al. 2023) hypothesize $p_\theta(\mathbf{z}_t|P_{\text{full}}) \propto p_\theta(\mathbf{z}_t | \emptyset) \prod_{i=1}^{N_c} \left(\frac{p_\theta(\mathbf{z}_t|P_i)}{p_\theta(\mathbf{z}_t|\emptyset)} \right)^{g_i}$, effectively approximating a complex distribution as the combination of simpler ones. Here, $\emptyset$ denotes the ‘‘null’’ condition, and g_i serves as the guidance factor of P_i . Therefore, the ‘‘Compose’’ operator used in these methods can be defined by applying the following score function in each time step:

$$\begin{aligned} s_t^\theta(\mathbf{z}_t | P_{\text{full}}) &= \nabla_{\mathbf{z}_t} \log(p_\theta(\mathbf{z}_t|P_{\text{full}})) \\ &\propto \nabla_{\mathbf{z}_t} \log(p_\theta(\mathbf{z}_t | \emptyset)) + \sum_{i=1}^{N_c} g_i \left(\nabla_{\mathbf{z}_t} \log(p_\theta(\mathbf{z}_t|P_i)) \right. \\ &\quad \left. - \nabla_{\mathbf{z}_t} \log(p_\theta(\mathbf{z}_t | \emptyset)) \right) \\ &\propto s_t^\theta(\mathbf{z}_t | \emptyset) + \sum_{i=1}^{N_c} g_i \left(s_t^\theta(\mathbf{z}_t | P_i) - s_t^\theta(\mathbf{z}_t | \emptyset) \right). \end{aligned} \quad (2)$$

Notably, Eq. (2) can be regarded as a multi-condition extension of Classifier-Free Guidance (Ho and Salimans 2022). In the following, we elaborate on the proposed Infer&Fuse, a more efficient DtC framework, detailing its implementations of the ‘‘Decompose’’ and ‘‘Compose’’ operators.

The ‘‘Decompose’’ Operator of Infer&Fuse

First, we introduce the ‘‘Decompose’’ operator of our Infer&Fuse. Given that pre-trained T2I models (Podell et al. 2023; Labs 2024) excel in processing simple text inputs, we initially employ a powerful LLM (Yang et al. 2025) to decompose the long text P_{full} into several sub-prompts. Specifically, each sub-prompt only contains a non-overlapping subset of semantic information indicated in P_{full} . To capture the dependencies among these sub-prompts, we also query the LLM to obtain the concise summary of P_{full} . Then, the summary is prepended to each sub-prompt, thereby deriving the representation of P_i . The overall pipeline is illustrated in Fig. 2.

The ‘‘Compose’’ Operator of Infer&Fuse

Next, we introduce the details of ‘‘Compose’’ operator used in Infer&Fuse. The challenge is to find the optimal guidance

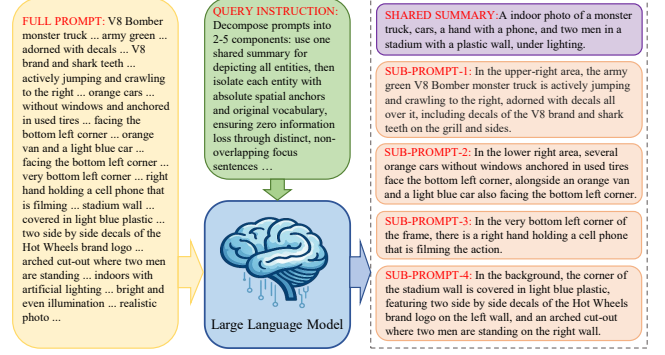


Figure 2: The ‘‘Decompose’’ operator used in our Infer&Fuse.

factors $\{g_i\}_{i=1}^{N_c}$ in Eq. (2). However, employing a global factor for each P_i is inefficient, since the contribution of each semantic element varies across different regions of an image. Furthermore, Eq. (2) fails to incorporate the inherent global context of P_{full} , leading to visual incoherence and the omission of semantic details. To address this, we extend the score function defined in Eq. (2) to:

$$\begin{aligned} s_t^\theta(\mathbf{z}_t | P_{\text{full}}) &\propto s_t^\theta(\mathbf{z}_t | \emptyset) \\ &+ \gamma \left((\mathbf{I} - \alpha_t) \cdot \left(\sum_{i=1}^{N_c} \omega_{t,i} (s_t^\theta(\mathbf{z}_t | P_i) - s_t^\theta(\mathbf{z}_t | \emptyset)) \right) \right. \\ &\quad \left. + \alpha_t \cdot (s_t^\theta(\mathbf{z}_t | P_{\text{full}}) - s_t^\theta(\mathbf{z}_t | \emptyset)) \right), \end{aligned} \quad (3)$$

where the hyperparameter γ adjusts the overall guidance scale. In particular, $\omega_{t,i}$ represents a time-dependent normalized weighting map with the shape (H, W) , satisfying $\sum_{i=1}^{N_c} \omega_{t,i}^u = 1$, $u \in [1, HW]$, where $\omega_{t,i}^u$ is the element of $\omega_{t,i}$ in the u -indexed pixel. Therefore, $\omega_{t,i}$ converts the original g_i to a spatially-varying, pixel-wise *guidance map*. Besides, we incorporate the global information through combining the guidance from $\{P_i\}_{i=1}^{N_c}$ and P_{full} using the time-dependent *blending map* α_t , enhancing the visual coherency.

Consequently, Eq. (3) is used to define the ‘‘Compose’’ operator of Infer&Fuse, which is illustrated in Fig. 3. Next, we introduce how to derive the $\{\omega_{t,i}\}_{i=1}^{N_c}$ and α_t in Eq. (3).

Derivation of $\{\omega_{t,i}\}_{i=1}^{N_c}$. Intuitively, given the current noisy latent $\mathbf{z}_t$ in the timestep t , the optimal $\{\omega_{t,i}\}_{i=1}^{N_c}$ is required to maximize the posterior probability $p_\theta(P_{\text{full}} | \mathbf{z}_t')$, where $\mathbf{z}_t'$ represents the next-step latent derived by Eq. (3). We first apply the first-order Taylor expansion and get the following objective:

$$\begin{aligned} \max_{\{\omega_{t,i}\}_{i=1}^{N_c}} \log(p_\theta(P_{\text{full}} | \mathbf{z}_t')) &\approx \log(p_\theta(P_{\text{full}} | \mathbf{z}_t)) + \\ &\sum_{u=1}^{HW} \underbrace{\nabla_{\mathbf{z}_t^u} \log(p_\theta(P_{\text{full}} | \mathbf{z}_t)) \cdot (\mathbf{z}_t'^u - \mathbf{z}_t^u)}_{\Delta_t^u}. \end{aligned} \quad (4)$$

Toy American Airlines jet, suspended and flying right, starting to cross the larger red planet. a smaller yellow one in the upper left part of the image and a larger red one on the bottom right. The background is a cosmic scene with a black, star-speckled sky, featuring sparkling stars and swirling cosmic dust around them ...

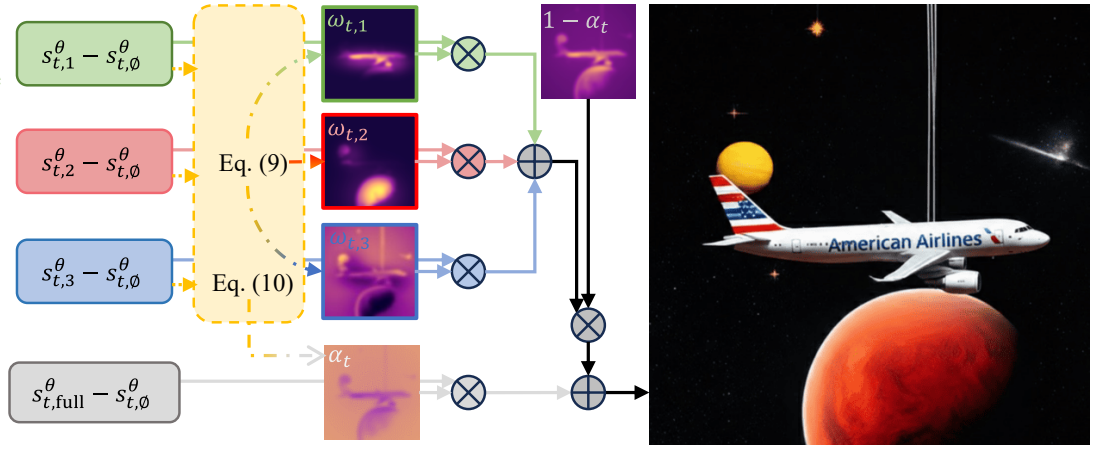


Figure 3: The pipeline of the “Compose” operator used in Infer&Fuse. We use different colors to denote each sub-prompt. As illustrated in the figure, we first compute weighting maps $\{\omega_{t,i}\}_{i=1}^{N_c}$ via Eq. (9) to combine the scores derived from each sub-prompt. Subsequently, we calculate a blending map α_t using Eq. (10), integrating the global guidance derived from P_{full} into the regions lacking sufficient semantic guidance.

We can further express Δ_t^u in Eq. (4) as:

$$\begin{aligned} \Delta_t^u &= \nabla_{\mathbf{z}_t^u} \log(p_\theta(P_{\text{full}} | \mathbf{z}_t)) \cdot (\mathbf{z}_t'^{i,u} - \mathbf{z}_t^u) \\ &\propto \sum_{i=1}^{N_c} \omega_{t,i}^u \nabla_{\mathbf{z}_t^u} \log(p_\theta(P_i | \mathbf{z}_t)) \cdot \underbrace{(\mathbf{z}_t^{i,u} - \mathbf{z}_t^u)}_{\Delta \mathbf{z}_t^{i,u}}, \\ &\quad \text{where } \mathbf{z}_t^i = \mathbf{z}_t + \eta_t \left(\mathbf{s}_t^\theta(\mathbf{z}_t | P_i) - \mathbf{s}_t^\theta(\mathbf{z}_t | \emptyset) \right), \end{aligned} \quad (5)$$

where η_t is specified by the used scheduler, and $\mathbf{z}_t^i$ denotes the next-step latent derived from the current $\mathbf{z}_t$ conditioned solely on P_i . More details of Eq. (5) are provided in the appendix. Consequently, solving Eq. (4) is equivalent to maximizing Eq. (5) independently at each pixel. Therefore, the optimal weighting factors $\{\omega_{t,i}^u\}_{i=1}^{N_c}$ in u -indexed pixel can be calculated by solving:

$$\begin{aligned} \max_{\{\omega_{t,i}^u\}_{i=1}^{N_c}} L^u &= \Delta_t^u + \underbrace{(-\tau \sum_i \omega_{t,i}^u \log(\omega_{t,i}^u))}_{L_e^u} \\ &\propto \sum_{i=1}^{N_c} \omega_{t,i}^u \nabla_{\mathbf{z}_t^u} \log(p_\theta(P_i | \mathbf{z}_t)) \cdot \Delta \mathbf{z}_t^{i,u} + L_e^u \\ &\propto \underbrace{\sum_{i=1}^{N_c} \omega_{t,i}^u \nabla_{\mathbf{z}_t^u} \log(p_\theta(\mathbf{z}_t | P_i) / p_\theta(\mathbf{z}_t | \emptyset)) \cdot \Delta \mathbf{z}_t^{i,u}}_{L_p^u} + L_e^u, \end{aligned} \quad (6)$$

where the additional L_e^u term regards the weighting factors as a probability distribution ($\sum_{i=1}^{N_c} \omega_{t,i}^u = 1$) and maximizes its entropy with $\tau > 0$. In essence, L_e^u prevents any pixel from being overly dominated by specific P_i . We further express

L_p^u in Eq. (6) as:

$$\begin{aligned} L_p^u &= \sum_{i=1}^{N_c} \omega_{t,i}^u \nabla_{\mathbf{z}_t^u} \log(p_\theta(\mathbf{z}_t | P_i) / p_\theta(\mathbf{z}_t | \emptyset)) \cdot \Delta \mathbf{z}_t^{i,u} \\ &= \sum_{i=1}^{N_c} \omega_{t,i}^u (s_t^{\theta,u}(\mathbf{z}_t | P_i) - s_t^{\theta,u}(\mathbf{z}_t | \emptyset)) \cdot (\mathbf{z}_t^{i,u} - \mathbf{z}_t^u) \\ &= \eta_t \sum_{i=1}^{N_c} \omega_{t,i}^u \underbrace{\|s_t^{\theta,u}(\mathbf{z}_t | P_i) - s_t^{\theta,u}(\mathbf{z}_t | \emptyset)\|_2^2}_{d_{t,i}^u}. \end{aligned} \quad (7)$$

By substituting Eq. (7) into Eq. (6), we have:

$$\max_{\{\omega_{t,i}^u\}_{i=1}^{N_c}} L^u = \eta_t \sum_{i=1}^{N_c} \omega_{t,i}^u d_{t,i}^u + (-\tau \sum_i \omega_{t,i}^u \log(\omega_{t,i}^u)). \quad (8)$$

Now, we can derive the weighting map $\{\omega_{t,i}^u\}_{i=1}^{N_c}$.

Proposition 1 (The solution of Eq. (8)) When $\omega_{t,i}^u$ takes the following form, $\{\omega_{t,i}^u\}_{i=1}^{N_c}$ is the solution of Eq. (8).

$$\omega_{t,i}^u = \frac{\exp(d_{t,i}^u / \tau)}{\sum_{j=1}^{N_c} \exp(d_{t,j}^u / \tau)}, \text{ where } d_{t,i}^u \text{ is defined in (7)}. \quad (9)$$

We provide the full derivation in the appendix. As shown in Fig. 3, the weighting maps highlight the dominant regions for the corresponding semantics, enabling distinct concepts to be generated at appropriate locations. Therefore, the spatially-varying guidance map employed in Infer&Fuse facilitates efficient composition.

Derivation of α_t . So far, we have derived the weighting maps $\{\omega_{t,i}^u\}_{i=1}^{N_c}$ in Eq. (3). However, the guidance from $\{P_i\}_{i=1}^{N_c}$ fails to model the global information, such as the

intricate dependencies among the semantic elements, impairing the visual coherence. To mitigate this issue, Infer&Fuse introduces a time-dependent blending map α_t to incorporate the global guidance from P_{full} , which contains the complete context. Formally, the element of α_t in the u -indexed pixel can be expressed as:

$$\alpha_t^u = \frac{\lambda_{\text{full}}}{\lambda_{\text{full}} + \sum_{i=1}^{N_c} d_{t,i}^u}, \quad (10)$$

where the hyperparameter λ_{full} controls the basic strength. Intuitively, Eq. (10) amplifies the global guidance in regions with insufficient semantic guidance ($\sum_{i=1}^{N_c} d_{t,i}^u$ is small), thereby complementing the missing information.

Overall Pipeline. As indicated in previous studies (Hertz et al. 2022; Brack et al. 2024), the early denoising steps construct the semantic structure, while the later steps focus on visual details. Therefore, Infer&Fuse only performs “Compose” operation in early steps, avoiding significant computational overhead during inference. In later stages, Infer&Fuse applies Classifier-Free Guidance (Ho and Salimans 2022) conditioned solely on P_{full} .

Experiments

Experimental Setup

Implementation details. We implement Infer&Fuse upon multiple T2I models. Here, we provide the results of Stable Diffusion 1.4 (Rombach et al. 2022) and 3.5 (Esser et al. 2024) (SD 1.4 & 3.5). For “Decomposition”, we adopt Qwen3.5 (Yang et al. 2025) as the LLM. More details and results are provided in the appendix.

Baselines. We compare Infer&Fuse with previous training-free approaches designed for the LT2I task. For optimization-based methods, we choose Attend&Excite (A&E) (Chefer et al. 2023), InfSplign (Rastegar et al. 2025), and Predicated Diffusion (Pred-Diff) (Sueyoshi and Matsubara 2024). RPG (Yang et al. 2024) is selected as the DtC-based method. We also evaluate methods under other paradigms, like SCoPE (Saichandran et al. 2025).

Benchmarks. We use two public benchmarks designed for the LT2I task: DPG-Bench (Hu et al. 2024) and DetailMaster (Jiao et al. 2025). They assess the model’s capability across multiple dimensions: *Entity/Object Presence* verifies successfully generated objects; *Attribute/Character Attributes* measures whether objects’ attributes align with the prompt; *Relation/Character Locations* checks (non-)spatial relation correctness; and *Global/Scene Attributes* evaluates overall coherence. DPG-Bench also uses the *Other* metric to indicate the counting accuracy and text rendering fidelity.

Evaluation and Analysis

As shown in Tab. 1-Tab. 4, Infer&Fuse consistently enhances the underlying T2I models across multiple dimensions.

Most optimization-based methods (Chefer et al. 2023; Rastegar et al. 2025) exhibit limited performance improvements, particularly for attribute metrics (*Attribute/Character Attributes* and *Relation/Character Locations*). The reason is

Model	Entity	Attribute	Relation	Global	Other
SD1.4	71.53	72.33	79.44	80.55	56.00
A&E	71.98	72.34	79.33	81.16	54.40
InfSplign	71.32	71.59	80.22	81.76	52.80
Pred-Diff	74.54	<u>74.31</u>	<u>82.54</u>	83.28	54.80
RPG	66.40	65.08	74.60	80.24	47.60
ScoPE	61.88	63.71	69.09	81.16	52.40
Ours	<u>73.55</u>	75.02	83.13	<u>82.83</u>	<u>55.80</u>

Table 1: The SD1.4 results on the DPG-Bench.

Model	Entity	Attribute	Relation	Global	Other
SD3.5	89.86	88.17	89.16	<u>92.33</u>	<u>89.19</u>
A&E	90.81	<u>89.35</u>	89.26	83.18	87.95
InfSplign	86.48	85.22	87.69	82.75	80.59
Pred-Diff	88.81	89.17	88.57	92.58	86.84
RPG	87.37	86.37	<u>89.28</u>	75.19	86.83
ScoPE	77.66	79.43	<u>82.16</u>	79.40	77.00
Ours	<u>90.51</u>	90.17	91.33	83.86	89.45

Table 2: The SD3.5 results on the DPG-Bench.

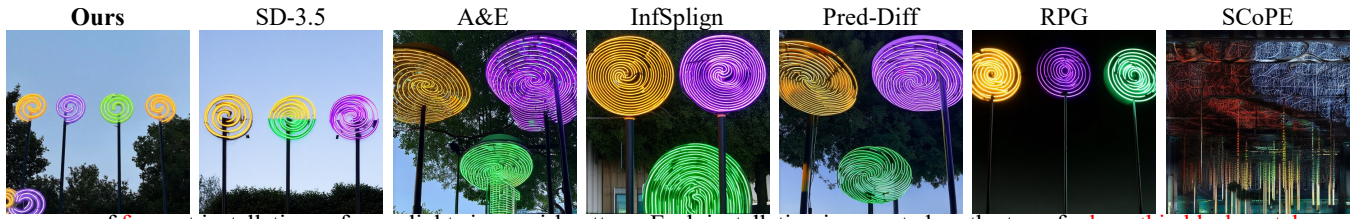
that optimizing attention scores is insufficient for enhancing non-entity concepts. As evident in Fig. 4, they fail to generate accurate layouts (wrong arrangement of neon lights and horses in 1st and 2nd rows). The DtC-based method RPG (Yang et al. 2024) also demonstrates the limited capability due to inefficient DtC implementation.

Our method achieves the best performance across multiple metrics on various benchmarks. As depicted in Fig. 4, Infer&Fuse exhibits superior capability in handling complicated (non-)spatial relations, while achieving accurate attribute alignment (e.g., object count and color). As shown in the 1st row of Fig. 4, only our method generates neon lights with accurate colors at the designated positions, while generating the bottom-left one that is neglected by other methods. Furthermore, our method also preserves most of the details depicted in the input. For the example indicated in the 4th row of Fig. 4, our method generates the scattered gold letters lying on the countertop, which are ignored by other methods.

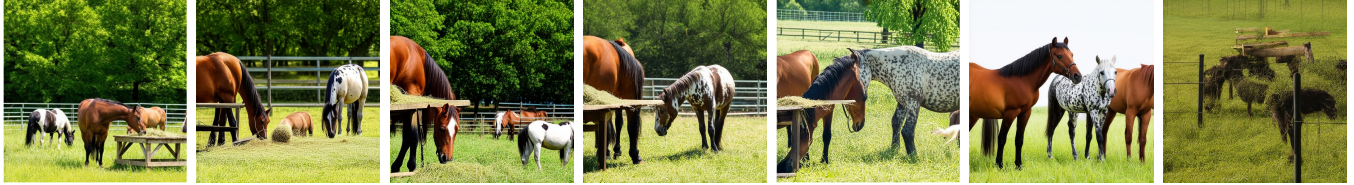
Ablation Studies

Tab. 5 assesses our key designs. First, we compare the weighting mechanism defined in Eq. (9) with the following strategies. The **Global** scheme defines a single factor for each sub-prompt as shown in Eq. (2). **Mean** and **Max** strategies apply “mean” and “max” operations in Eq. (9).

As shown in Tab. 5, our method outperforms all these settings across multiple metrics. Limited by employing a single guidance factor across the entire image, the **Global** strategy suffers from object omission and attribute confusion. As depicted in the 1st row of Fig. 5, it fails to generate the owl. The **Mean** operation suffers from the dilution of guidance from dominant sub-prompts, resulting in ambiguity. As shown in the 1st row of Fig. 5, the guidance derived from the “white cat” and “owl” sub-prompts are comparable in the owl region,



...a row of **four** art installations of neon lights in a swirl pattern. Each installation is mounted on the top of a **long thin black metal pole**. The two outside swirl lights are **yellow and orange** neon swirl lights, while the inside swirl lights are **purple** neon swirl lights and a **green** neon swirl light ...The **bottom** of the background is lined with bushy **treetops** and the **top half** is **light blue clear skies**. ...The **purple and yellow** neon light art installation **partially visible** is positioned in the **lower left corner**, separate from the main row of lights.



Four horses are in a grassy... A **brown** horse, with the bottom of its legs being **black**, is turning its head to **eat hay from a wooden table**. A **white and black spotted** horse is **behind**... A **lighter brown** horse... The snout of a horse is **partially cut from the right side** of the image... Image is brightly lit with sunlight... The brown horse and the white and black spotted horse are **separated by a distance**...



... **five black statues**. Two statues are standing on a large **gray rock**... The **woman** is wearing a **large robe** and is **holding onto a baby** that is **covered except for its feet**. To the left of the statues is **another statue of a woman**... To the right of the **woman**... there is a **sheep**... Behind the statues, there is a **grass field that has multiple trees**... Between the trees, a **blue sky** and a small **white cloud**...



...wood panel wall with a **brown color**... A **neon light sign** of an **ice cream cone**, mounted in the **center**... A **dark green rubber plant** in a **red clay pot**, sitting on **top of a white countertop**... a small **stainless steel blender** with a **cylindrical shape, metallic color**... **letter board**... has **golden letters** that read... "**SMALL \$6/ TRIO \$8/ PINT \$12/ CONE 50¢**"... Various other **gold letters** lay on the **countertop in front of the upright letter board**...



... a **round coopered link basket** fruit bowl... **no fruit** inside of it... display card with the words "**Banana**", in **black text**. Underneath "**Banana**", are the words "Please remember to wash your fruits and / vegetables before cutting or eating", in a **much smaller black text**... **black containers** are **partially visible** with **white papers** inside them... a **gold colored container**...

Figure 4: Qualitative results on the DetailMaster (Jiao et al. 2025) benchmark. The results from our Infer&Fuse exhibit high fidelity in terms of complex attributes (e.g., object count and color) and (non-)spatial relations among the entities. In contrast, other methods demonstrate poor performance in semantic alignment.

confusing their appearances. In addition, the **Max** strategy mitigates this issue by selecting the dominant sub-prompt

for each region. However, as shown in the 2nd row of Fig. 5, this leads to inferior image quality. In contrast, our weighting

Model	Object	Character Attributes			Character	Scene Attributes			
	Presence	Object	Animal	Person	Locations	Background	Light	Style	Spatial
SD1.4	18.36	10.50	10.47	2.81	6.64	19.51	47.93	60.04	2.09
A&E	18.19	10.81	8.96	3.09	6.70	19.10	48.38	58.07	1.47
InfSplign	20.47	12.74	12.37	1.99	6.88	18.56	50.56	62.06	1.97
Pred-Diff	20.76	<u>12.89</u>	<u>14.55</u>	5.94	<u>9.07</u>	<u>22.39</u>	<u>54.35</u>	58.93	2.15
RPG	20.73	10.25	16.14	4.28	7.33	16.33	49.00	61.94	<u>2.24</u>
ScoPE	15.64	8.81	8.46	3.16	4.98	11.65	40.47	58.98	1.34
Ours	21.43	12.90	14.09	<u>4.32</u>	9.86	32.71	60.07	64.72	2.95

Table 3: The SD1.4 results on the DetailMaster (Jiao et al. 2025) benchmark.

Model	Object	Character Attributes			Character	Scene Attributes			
	Presence	Object	Animal	Person	Locations	Background	Light	Style	Spatial
SD3.5	43.11	45.58	<u>42.80</u>	35.94	33.45	<u>83.17</u>	<u>83.76</u>	86.70	9.68
A&E	42.61	40.72	40.30	22.71	31.45	71.99	67.09	59.66	8.56
InfSplign	<u>45.17</u>	40.86	41.96	22.32	32.25	76.00	79.98	83.56	8.80
Pred-Diff	39.75	38.21	42.36	41.97	28.12	68.97	70.96	75.04	7.18
RPG	36.75	34.60	40.29	35.53	27.07	37.93	67.87	78.58	6.65
ScoPE	20.96	23.25	33.36	14.53	15.27	21.98	53.67	55.78	2.31
Ours	49.70	<u>44.76</u>	43.33	<u>36.46</u>	40.07	85.43	87.50	89.65	11.65

Table 4: The SD3.5 results on the DetailMaster benchmark.

Model	Object	Character Attributes			Character	Scene Attributes			
	Presence	Object	Animal	Person	Locations	Background	Light	Style	Spatial
Global	43.11	<u>45.58</u>	42.80	<u>35.94</u>	33.45	83.17	83.76	<u>86.70</u>	9.68
Mean	38.07	39.44	42.80	23.53	28.08	71.87	84.12	84.92	7.98
Max	<u>44.16</u>	46.43	43.18	34.23	<u>35.34</u>	<u>83.98</u>	<u>85.84</u>	86.17	<u>10.17</u>
w/o α_t	41.64	42.75	44.40	30.55	31.43	80.09	84.40	85.29	9.35
Ours	49.70	44.76	<u>43.33</u>	36.46	40.07	85.43	87.50	89.65	11.65

Table 5: Ablation studies on DetailMaster benchmark using SD3.5.

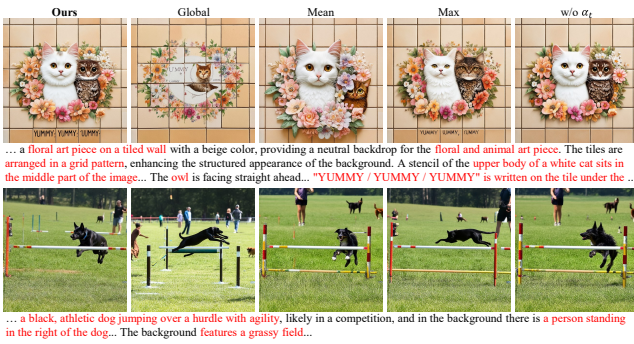


Figure 5: Ablation studies on the DetailMaster benchmark.

mechanism yields superior visual quality while maintaining high semantic alignment with the input text.

We also provide the results from the method without the blending map (w/o α_t), where the α_t in Eq. (3) is set to

an all-zero matrix. As shown in the 2nd row of Fig. 5, the absence of global information prevents the model from accurately modeling relationships between distinct entities, where the person is placed on the left side of the dog by mistake. Furthermore, as depicted in the 1st row of Fig. 5, it suffers from missing details, where the designated text is missing. Tab. 5 further demonstrates that the incorporation of global information can enhance the semantic alignment.

Conclusion

Existing T2I methods exhibit poor performance in LT2I generation. Therefore, we propose Infer&Fuse, a DiC framework designed for enhancing the LT2I generation capabilities. It decomposes long text into several sub-prompts and combines the derived scores with efficient weighting mechanism. In addition, it incorporates the global guidance for enhancing overall visual coherence. Extensive experiments demonstrate that various T2I models equipped with our Infer&Fuse achieve consistent performance gains on multiple benchmarks.

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